

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.4 TRIGONOMETRY 1: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance, and how does this help Kenyans solve real problems in construction, surveying and everyday life? Answer every section in the space below it. Use the Kenya Power mast near the school in Nakuru County as your main example: a student stands 40 m from the base of the mast and measures the angle of elevation to the top as about 50° . Draw neat, labelled diagrams, show every step of your working, include units, and write a full sentence for each final answer. You may use your mathematical tables, calculator and Special Angles Reference Card. Where a table value is needed it is given to you.

Part 1: The missing tool

On Day 1 you were asked whether you could find the height of the Nakuru mast using only a horizontal distance of 40 m and an angle of about 50° . (a) Draw the right-angled triangle for this situation. Mark the 50° angle and the 40 m side, and label the three sides as opposite, adjacent and hypotenuse relative to the 50° angle. (b) Explain why Pythagoras' theorem alone could not solve the problem. (c) In Lesson 1 you drew right-angled triangles of different sizes that all shared the same acute angle and calculated side ratios. What did you notice about the ratios, and why does this make a table of ratios useful for finding the mast's height? (d) Write the names and definitions (as ratios of sides) of sine, cosine and tangent.

Write your answer here:

Part 2: Solving the mast problem

Use the triangle from Part 1. (a) Which trigonometric ratio links the 50° angle, the 40 m side and the unknown height h ? Give a reason based on which sides are involved. (b) Write the equation, rearrange it to make h the subject, and calculate the height of the mast. Use $\tan 50^\circ \approx 1.192$. (c) Explain in a sentence how this answers the question from Day 1 without using ladders, drones or guessing.

Write your answer here:

Part 3: Relationships between the ratios

(a) In the mast triangle, work out the size of the other acute angle (at the top of the mast). Then compare $\sin$ of that angle with $\cos 50^\circ$. Use the triangle to explain what you find. Your explanation must refer to opposite and adjacent sides. State the general rule for any acute angle θ . (b) Suppose the $\tan$ button on a surveyor's calculator is broken. Which identity connecting $\tan \theta$ with $\sin \theta$ and $\cos \theta$ could she use, and why does it work? Show the short algebraic proof using the definitions of the ratios. (c) Given that $\sin \theta = 3/5$ in a right-angled triangle, find $\cos \theta$ and $\tan \theta$ without using a calculator.

Write your answer here:

Part 4: Special angles, elevation and depression

(a) Imagine the student, still 40 m from the base, measured an angle of elevation of exactly 45° instead of 50° . Using the exact value of $\tan 45^\circ$ from your Special Angles Reference Card, find the height of the mast and explain how you obtained the exact value of $\tan 45^\circ$ from a triangle. (b) Now imagine the angle of elevation was 60° . Write the exact height using the exact value of $\tan 60^\circ$, and give it to one decimal place. (c) With the original 50° angle of elevation, a worker stands at the top of the mast and looks down at the student on the ground 40 m from the base. What is the angle of depression of the student from the worker? Explain your reasoning, and say how an angle of depression differs from an angle of elevation.

Write your answer here:

Part 5: Using trigonometry in real life across Kenya

A wheelchair ramp is being built at Kisumu County Referral Hospital. It must rise at an angle of 12° to the ground and cover a horizontal distance of 6 m. (a) Draw and label the triangle. (b) Use the SOH-CAH-TOA decision steps to find the length of the ramp's sloping surface. Use $\cos 12^\circ \approx 0.9781$ and give your answer to two decimal places. (c) Describe the general steps you follow to solve any right-angled triangle problem. (d) Name two Kenyan professionals who could use these methods, say what each would measure, and explain how this answers the driving question.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Labelling and choosing the correct ratio	Draws accurate, fully labelled triangles with opposite, adjacent and hypotenuse correct relative to the given angle, and justifies every ratio choice using SOH-CAH-TOA.	Labels triangles mostly correctly and chooses the right ratio, but the justification is brief or incomplete.	Labels sides incorrectly or does not give a reason for choosing the ratio.
Calculation and use of tables, calculator or exact values	Sets up and rearranges equations correctly, calculates accurately (mast about 47.7 m, ramp about 6.13 m, exact special-angle values) with units and sensible rounding.	Sets up equations correctly with minor arithmetic, rounding or unit errors.	Equations are set up incorrectly or the calculation is mostly incorrect or incomplete.
Explaining relationships between ratios	Clearly explains $\sin \theta = \cos(90^\circ - \theta)$ using opposite/adjacent swapping and proves $\tan \theta = \sin \theta \div \cos \theta$ from the	States the relationships correctly but explanation or proof is partial.	States the relationships without explanation or with errors.

	definitions.		
Elevation, depression and connection to the phenomenon	Correctly distinguishes elevation from depression, explains why they are equal, and clearly links all work to how the Nakuru mast height is found without climbing.	Defines elevation and depression correctly with a weak link to the phenomenon.	Confuses elevation and depression or does not connect the work to the mast.
Real-life reasoning and communication	Gives a clear general method and relevant Kenyan examples, with well-organised working and full-sentence conclusions.	Gives a reasonable method and examples, with some gaps in organisation.	Gives little or unclear reference to real-life use and working is disorganised.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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